



Policy Analytics Challenge

**| Effects of climate shocks on
educational outcomes,
an application to Brazil.**

Federico Veneri

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FOCUS OF THE PROJECT



OBJECTIVE

This project aims to explore the usefulness of public available dataset to:

- Quantify the effect of extreme climate condition on developmental outcomes (health/education).
- Provide some examples of modeling strategies that could be used to quantify these effects.



DATA

I focus on Brazil for this project, but LAC countries have similar data**.

- Mortality data.
- Hospital admission.
- Education attendance.
- Education outcomes.
- Extreme weather events

For Brazil, I explored:

- Mortality: Datasus-SIM
- Hospital admission: Datasus-SIH
- **Educational outcomes:**
 - SAEB, pass rate



ANALYSIS & RESULTS

Data was aggregated to municipality level; climate shocks were defined as historical extreme rain events.

- **Education:** Extreme rainfall was associated with lower standardized test results on math and language, and lower pass rates at municipality level.
- **Health:** forthcoming.

Climate challenges and education.

- Between 2022-2024, 400 million students globally have experience school closure due to extreme weather ([WB,2024](#)).
 - Higher impacts on lower income countries: 18 days lost per year (10% of academic year)
 - Even if a school remains open, data shows reduced attendance under climate shocks.
 - Lost schooling translate to learning deficits. Not only days lost but also interruptions in pedagogy plans.
 - Learning deficits affects human capital accumulation and ultimately growth.
- Evidence from Latin America and the Caribbean, provides additional evidence in same direction.
 - [Bernal & Vlaicu \(2023\)](#): Colombia
Rural households re allocate children time from school to domestic activities when faced with climate shocks.
 - [Lima et. al. \(2024\)](#): Brazil
Extreme weather events (Rainfall) results in adverse impact on standardized test results.
- The previous results showcase recent results for LAC associated with rainfall, but the literature is more richer:
 - Other extreme weather events (heatwaves, droughts) have also been studied.
 - And other developmental outcomes (hospital admission, agricultural outcomes)
- My work is focus on rainfall shocks and educational challenges in Brazil
 - Tries to reproduce [Lima et. al. \(2024\)](#) based on public data.

Data and challenges

Municipal Mesh ([IBGE](#)).

- The **Brazilian Institute of Geography and Statistics** produce official shapefiles with Municipalities, UF, etc.
Acces: The shapifles are readily accesible, code implement to download the files. An R package is also available: geobr ([IPEA](#))

Education microdata ([INEP](#))

Sistema de Avaliação da Educação Básica (SAEB) and School Census (censo escolar).

- Provides infomation on educational outcomes (language and math) at school and student level.
 - Data available for every two years (1995-2023)
 - Controls can be created at indivdual ([SAEB](#)) or school level ([using school census](#)). Potential geocoded for schools.

Acces: The microdata is accesible, code implement to download the files.

- **Challenges:** The municipality and school code is hashed, but previous author have aquired the unhashed data.
 - Some public reports have municipality level aerages.

Climate

Copernicus ERA5-Land

- Provides information on key climate variables (e.g.: wind, precipitation, temperature).
 - Data is available hourly with a 9km resolution. Daily aggregations can be derived (sum , max, etc.)

Acces: Some R package allow to access the data, KrigR ([Kusch and Davy.2022](#)) on a predefined bounding box

- **Challenges:** The granularity of the data results in considerable large dataset.
 - Generating meaningfull agregations for municipalities is not straight foward.

Data and challenges

- While the previous slides provide the data used, additional sources were identified in Brazil.

Health Indicators-Datasus ([Ministry of health](#)) (FORTHCOMING)

Sistema de Informações de Mortalidade (SIM) ([Book](#), [variables](#))

- Is one of the oldest health data sources in Brazil (since 1975) unifying standardized data about death certificates.
- Indicator: mortality rates by age categories, causes and municipality (Admin2)

Sistema de Informações Hospitalares do SUS (SIH) ([Book](#), [variables](#))

- Provides administrative information (since 1981) about hospital admission, length of stay and cost.
- Indicator: hospital admission by age categories, causes and municipality (Admin2).

Cadastro Nacional de Estabelecimentos de Saúde ([Book](#), [variables](#))

- Provides additional information to create control variables like infrastructure, hospital personnel. Geocoded Hospitals.

Access:

- Several ways to access DATASUS, a package currently in development ([Microdatasus](#), [Saldanha et al](#)) has some functionalities.
- R function to download and perform primary cleaning implemented (Download_SIM_files, Download_SIH_files).

Challenges: server calls* are restricted for IP outside of Brazil. A VPN and several tries can help.

Municipality level variables

Sistema IBGE de Recuperação Automática ([SIDRA](#))

Access: The IBGE has made available an API to download the data. The R package: [sidrar](#), interface to access the data.

Challenges: Requires the table index. Few variables for municipalities

- I use population to showcase some back of the envelope calculations.

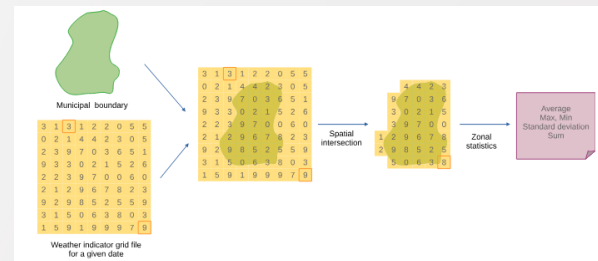
Feature engineering

Education outcome- Sistema de Avaliação da Educação Básica (SAEB) .

- I focus on yearly Municipality data for the 9th grade (last year of mandatory basic education)
- Considering:
 - Test scores (math and language): a numerical score derived from a standardized test.
 - Passing rate: Number of student who pass / Number of students
- Note: Scores are “comparable” between year, but to ensure interpretation I standardized the yearly scores for SAEB in math and language.

Rainfall shocks:

- I focused on zonal statistic indicators derived from RA5-Land ([Saldanha et al.2024](#))
- To generate meaningful aggregations the authors suggest
 - Consider the areal overlap
 - Compute weighted operations based on percentage of area overlapping.
 - The [authors](#) made available daily municipality level data (1950-2022)
- For each m municipality, I aggregated rainfall to monthly precipitation (1980-2022).
- Computed the historical mean and standard deviation (μ_{jm}, σ_{jm})
 - For each j month, consider: $S_{mtj}^k = 1 \{rainfall_{mjt} > \mu_{jm} + k \sigma_{jm}\}$
- We compute $NS_{mt}^k = \sum_j S_{mtj}^k$ and define that the **municipality experience a shock** if $S_{mt}^k = 1 \{NS_{mt}^k \geq 1\}$
- A similar approach was previously used by [Lima et al \(2024\)](#).
 - Considering k standard derivations allows to examine the size of the shocks.



MODELING

Model 1:

- I exploited the exogenous climate shocks, their timing and geographical variation using the following specification:

$$y_{mt} = \beta_0 + \beta_1 S_{mt}^k + \theta_m + \eta_t + \epsilon_{mt}$$

Where: y_{mt} denotes the outcomes of interest in municipality m at time t (passing rate, standardized test results in math and language)

S_{mt}^k denotes an indicator variable representing the rainfall shock, considering k standard deviations from the historical average.

θ_m and η_t denotes municipality and year fixed effects.

Note: Future work could consider school or student level information to consider better control variables and more policy relevant question.

Model 2:

- Our secondary specification:

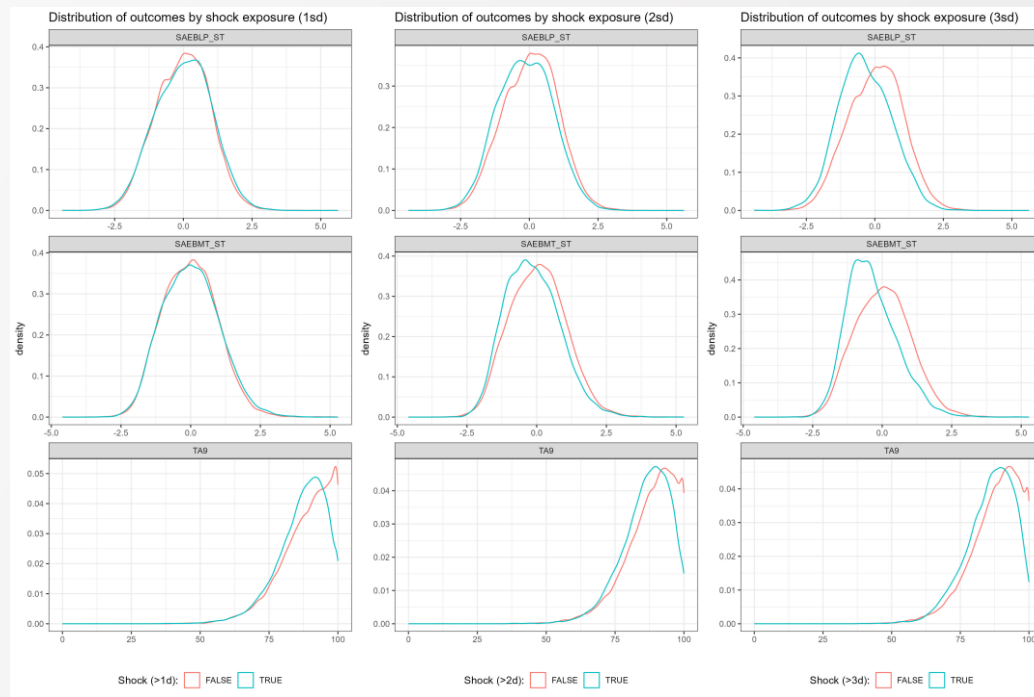
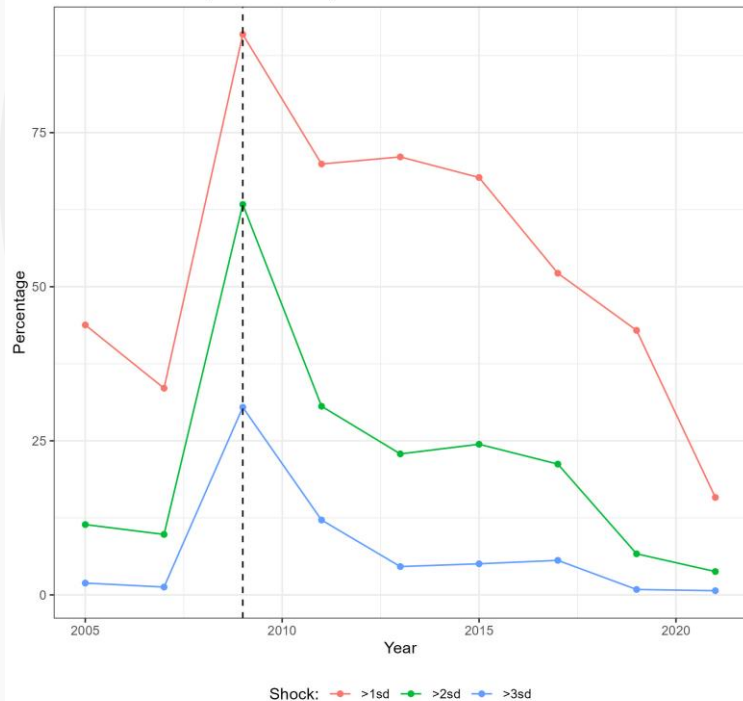
$$y_{mt} = \beta_0 + \beta_1 NS_{mt}^k + \theta_m + \eta_t + \epsilon_{mt}$$

Where: NS_{mt}^k denotes the number of months that can be consider extreme.

Exploiting the size of the shock and number of months that can be consider extreme allows to study heterogenous effects.

DESCRIPTIVE STATISTICS

Percent of municipalities that experience a climate shocks



- Municipalities facing larger shocks are associated with:
 - Lower passing rate
 - Lower average test results in math
 - Lower average test results in language

MODELING: RESULTS

	Rainfall Shock (1sd)			Rainfall Shock (2sd)			Rainfall Shock (3sd)		
	SAEB Math std	SAEB Language std	Pass (%)	SAEB Math	SAEB Language std	Pass (%)	SAEB Math std	SAEB Language std	Pass (%)
Rainfall Shock	-0.004 (0.006)	-0.001 (0.006)	-0.141* (0.068)	-0.046*** (0.007)	-0.030*** (0.007)	-0.383*** (0.083)	-0.091*** (0.010)	-0.068*** (0.011)	-0.284* (0.127)
FE: Year	Y	Y	Y	Y	Y	Y	Y	Y	Y
FE: Municipality	Y	Y	Y	Y	Y	Y	Y	Y	Y
Num.Obs.	47645	47645	49180	47645	47645	49180	47645	47645	49180
R2	0.737	0.701	0.546	0.738	0.701	0.546	0.738	0.701	0.546

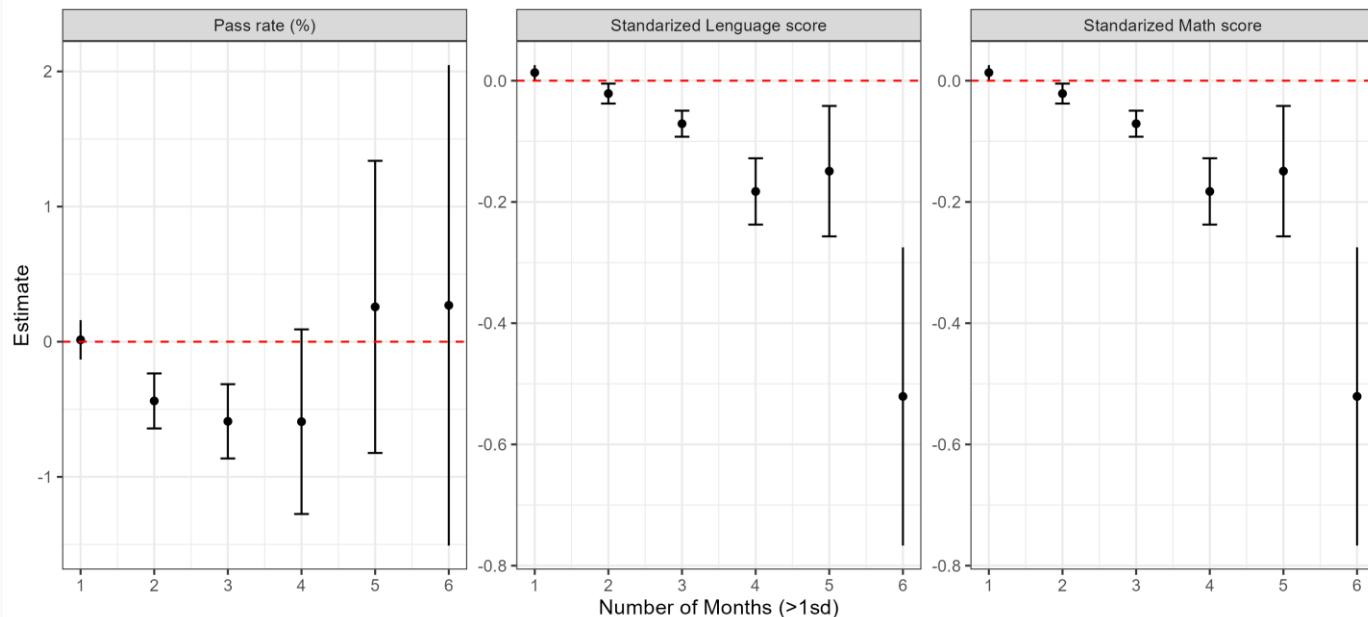
Note: + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

- Faced with a Rainfall shock, we can expect (statistical significantly):
 - Lower passing grades, even for lower shocks.
 - Lower average test results in both math and language.

Note: as a robustness check, a model without 2009 was also estimated. Results were qualitatively similar.

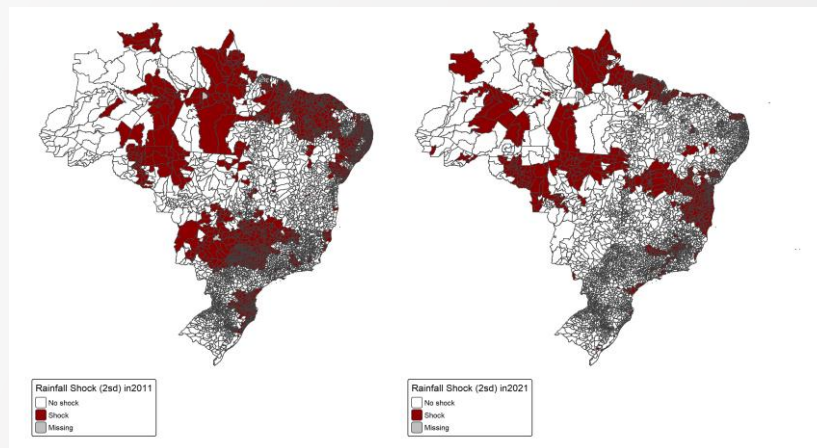
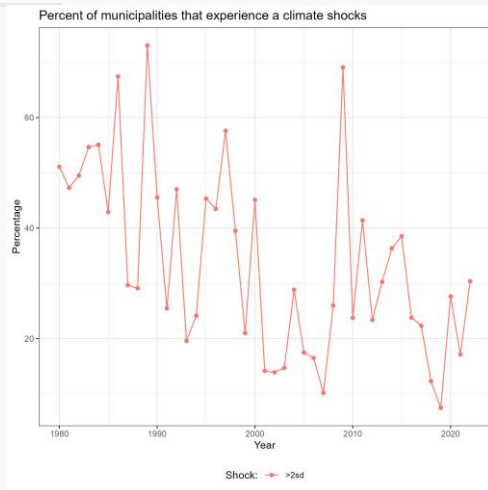
MODELING: RESULTS

Coefficient associated with number of shock
(Fixed Effect Models, Shock 1sd)



- Faced with more extreme weather month:
 - We observe a larger decrease in test average test result.
 - There seems to be a u-shape for passing rate.

CLIMATE CHALLENGES IN CONTEXT



- Rainfall shocks are not a one-time events for Brazilian municipalities.
- Municipalities that experiences this shocks are not always the same.
- The size of the shocks in terms of population is relevant, if we consider the cases of 2021:

Shocks (2Sd)	Population	% Population	Municipalities	% Municipalities
N	185.2 M	87%	4614	83%
Y	28.1M	13%	953	17%

POLICY IMPLICATIONS

Previous results suggest that a rainfall shocks are associated with:

- A decrease in pass rate and learning outcomes.
- Larger impact when more extreme (and sustained) rainfall is observed.
- Climate shocks affect human capital accumulation and can hinder future economic growth.

Study of the mechanism and heterogenous effects can help design effective policy.

- Are shock associated with lower attendance rate? Yes, ([WB.2024](#)).
 - If yes, remote learning plans can reduce learning losses and mitigate climate challenges.
 - Developing digital solutions for education is on the IADB's [focus area](#).
 - Before implementing any policy more information about access to remote learning is needed.
- Are shocks larger on communities/school with less infrastructure?
 - If yes, investment in infrastructure can help mitigate rainfall shocks.
 - However, this are more expensive and may require larger investments.

Study limitations:

- The current study uses municipality aggregated data, but micro data is available to inform our policies.
- Only one outcome was considered, study can be further expanded (heatwaves-hospital admission).
 - The study of other climate challenges, require more specialized modeling: Distributed lag (non-linear) models.



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